6 Months Training TR-104 Full Stack Development

March, 2026

Day 8 – Role-Based UI and Access Control

1. Objective of the Day

The objective of the day was to implement role-based access control and ensure that the user interface behaves according to user roles (admin and normal user).

2. Tasks Performed

- Implemented role-based access logic in the application
- Restricted admin panel access only to users with admin role
- Prevented normal users from accessing admin routes
- Controlled visibility of UI elements based on user role
- Displayed different navigation options for admin and users
- Integrated role-based checks with authentication service
- Tested access control using different user roles
- Verified secure navigation between panels

3. Tools / Technologies Used

- Angular Framework
- Angular Routing
- Firebase Authentication
- TypeScript

4. Key Learnings

- Role-based access control enhances application security

- UI should adapt based on user permissions
- Authentication and authorization must work together
- Services help manage user roles effectively
- Secure routing prevents unauthorized access

5. Challenges Faced

- Managing role-based logic across multiple components
- Ensuring users cannot bypass restricted routes

6. Solutions

- Used centralized authentication service for role management
- Implemented route guards to restrict unauthorized access

7. Conclusion

- Successfully implemented role-based UI and access control
- Ensured secure and role-specific navigation in the application